Class #10 - Orthogonality

1. Def: Let u and v be vectors in $\mathbf{R^n}$.
   The product $u^T v$ is called the *scalar product* or _____ of u and v (It is often written as $u \cdot v$).

2. Theorem: Let u, v, and w be vectors in $\mathbf{R^n}$, and let c be a scalar. Then
   (a) $u \cdot v =$ _____
   (b) $(u+v) \cdot w =$ _____
   (c) $(cu) \cdot v =$ _____ $=$ _____
   (d) $u \cdot u \geq$ _____, and $u \cdot u = 0$ if and only if _____.

3. Def: The *length* (or *norm*) of v is the _____ scalar $||v||$ defined by $||v|| =$ _____ $=$ _____, and $||v||^2 =$ _____.
   Note: $||cv|| =$ _____ for any scalar c.

4. Def: For u and v in $\mathbf{R^n}$, the *distance between u and v*, written as $\text{dist}(u, v)$, is the length of the vector $u - v$. That is,
   $$\text{dist}(u, v) = ||u - v||.$$

5. For two *nonzero* vectors u and v in $\mathbf{R^2}$ or in $\mathbf{R^3}$,
   $u \cdot v =$ _____
   where θ is the angle between u and v, _____.

6. Two *nonzero* vectors are _____ if the angle between them is $\theta = \pi/2$. $\Rightarrow$ _____ $= 0$.
   We generalize this fact for any vectors (even for zero vectors) in $\mathbf{R^n}$.

7. Def: two vectors u and v in $\mathbf{R^n}$ are *orthogonal* (to each other) if $u \cdot v = 0$.
   Note: _____ is orthogonal to every vector in $\mathbf{R^n}$.

① $u \in \mathbb{R}^n$ $v \in \mathbb{R}^n$

scalar / matrix / vector

ⓐ uv^T is a matrix.

ⓑ u^Tv is a scalar.

② $(A+B)^T = A^T + B^T$

③ $(cA)^T = cA^T$

↗ scalar

④



$d = \sqrt{(x_1-x_0)^2 + (y_1-y_0)^2}$

$\mathbb{R}^n$

(Q#5 Mm PP 11-13)

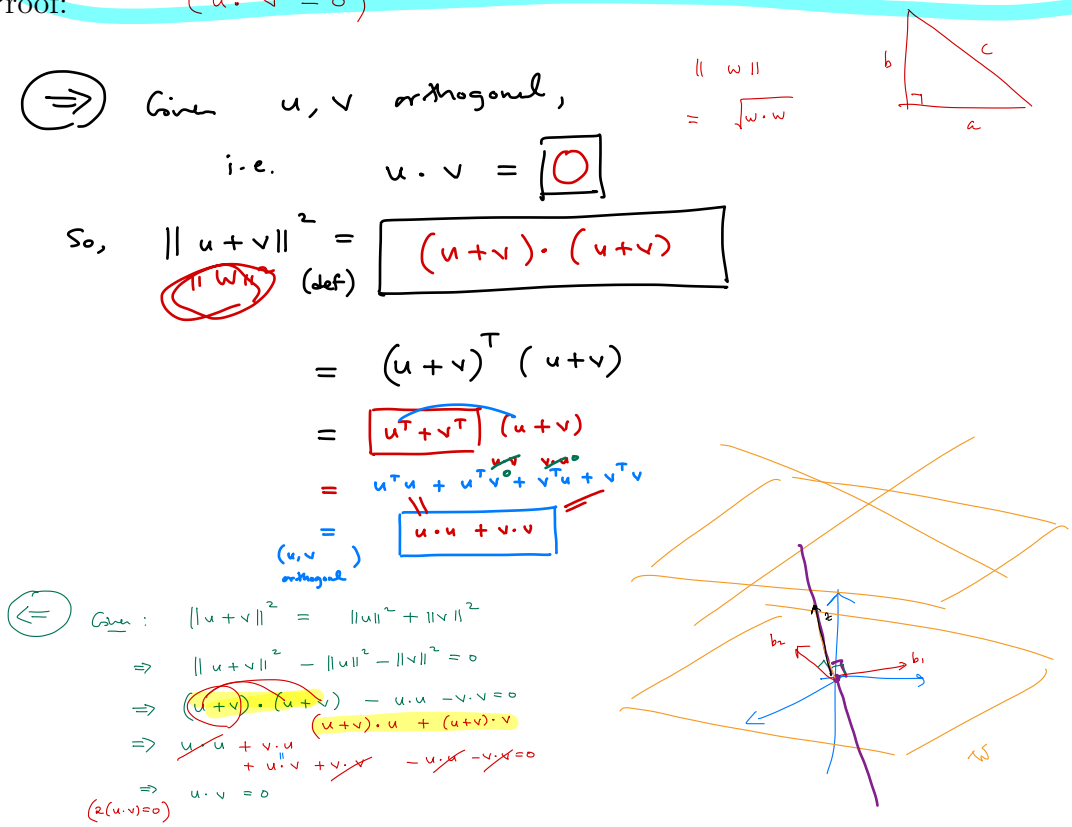
Midterm 10/26 Mm.

upto 10/23 lecture

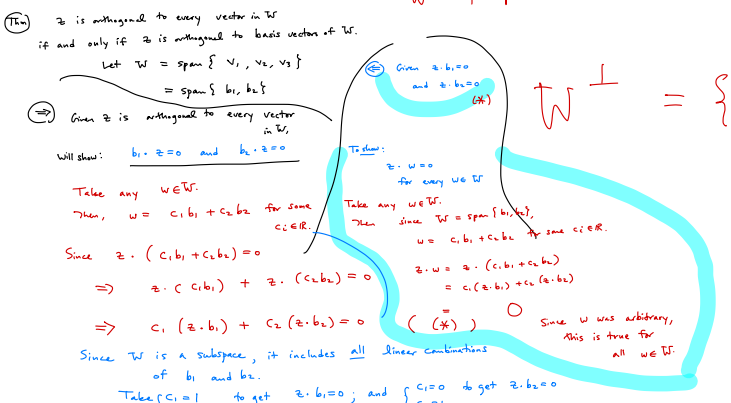
Q#4 Avg.

8.29

8. The Pythagorean Theorem:
Two vectors u and v are orthogonal if and only if $||u+v||^2 =$ _____ + _____.
Proof:



9. Def: Let W be a subspace of $\mathbf{R^n}$. If a vector z is orthogonal to every vector in W, then z is said to be *orthogonal to* W.
The set of all vecters z that are orthogonal to W is called the *orthogonal complement of* W and is denoted by $W^{\perp}$.



10. Examples: Suppose $n \geq 2$.
If $W = \{0\}$, then $W^{\perp} =$ _____
If $\dim W = 1$, then $W =$ _____, and there exists a nonzero vector x in $\mathbf{R^n}$ such that _____.
Then x is orthogonal to _____ of W.

Fill in the blank: A vector $x \in W^\perp$ if and only if x is orthogonal to every vector in a set that spans W.

$W = \text{span}\{\text{Spanning set}\}$ // (Spanning set)

(proof in the previous page)

If $\dim W = n$, then $W^\perp = \{0\}$

$0 \cdot w = 0 \ \forall w \in W$

$\{0\} \subset \mathbb{R}^n$ $\quad \{z \in \mathbb{R}^n \mid z \cdot w = 0 \ \forall w \in W\}$

11. Theorem: Let W be a subspace of $\mathbf{R^n}$. Then $W^\perp$ is a subspace of $\mathbf{R^n}$.

Proof:

(a) $u \in W^\perp$, $v \in W^\perp$ $\Rightarrow$ $u+v \in W^\perp$ (pp)

(b) $u \in W^\perp$, c scalar $\Rightarrow$ $cu \in W^\perp$

$m \begin{bmatrix} & n & \end{bmatrix}$

12. Example: Let A be an $m \times n$ matrix. Then, $(\text{Nul}A)^{\perp} =$ _____.

$\text{Nul}\,A = \{ x \in \mathbb{R}^n \mid Ax = 0 \}$ is a subspace of $\mathbb{R}^n$.

Avg. 8.29

Curve

(a) $A = \begin{bmatrix} 0 & 1.6 \\ 0.3 & 0.8 \end{bmatrix}$ (5 pts)

(b) $\lambda = 1.2$ (5 pts)

population growing $= 0.4$